Project Question / Problem Statement

“Telecommunications Customer Churn Prediction Using Machine Learning”

A telecommunications company provides internet, mobile, and bundled subscription services to thousands of customers. Recently, the company has experienced an increase in customer churn, resulting in revenue loss as customers cancel their subscriptions and switch to competitors.

The objective of this project is to develop a **machine learning-based Customer Churn Prediction System** that can predict whether a customer is likely to **churn (leave the service) or stay**.

The project should include the following tasks:

1. **Data Preprocessing**
    1. Handle missing and duplicate values.
    2. Convert categorical variables into numerical form.
    3. Detect and handle outliers where appropriate.
    4. Perform feature scaling/normalization where required.
    5. Split the dataset into training and testing sets.
2. **Exploratory Data Analysis & Visualization**
    1. Analyze customer demographics, subscription types, tenure, monthly charges, contract type, payment method, etc.
    2. Visualize the relationship between important features and churn.
3. **Handle Class Imbalance**
    1. Check whether the churn and non-churn classes are imbalanced.
    2. If the dataset is imbalanced, apply an appropriate balancing technique such as **SMOTE Or Random Oversampling**
    3. Compare model performance before and after balancing.

Model Building

Build and evaluate two classification models:

1. **Logistic Regression**
2. **K-Nearest Neighbors (KNN)**

For Logistic Regression perform **Cross-Validation**.

Model Evaluation & Comparison

Compare Logistic Regression and KNN using:

1. Accuracy
2. Precision

3. Recall
4. F1-score
5. Confusion Matrix
6. Cross-validation score

Since identifying customers who are actually going to churn is important for the business, give particular attention to **Recall and F1-score**, rather than relying only on accuracy.

2. **Recommended Question?**

Business Recommendation:

Based on the customer churn prediction results, what preventive business strategies should the telecommunications company adopt to reduce customer churn and improve customer retention?

Final Research Question

"How effectively can Logistic Regression and K-Nearest Neighbors (KNN) predict customer churn in a telecommunications company, and which model performs better after appropriate data preprocessing, class-balancing techniques, and cross-validation?"